Raunaq Singh

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EDUCATION

Georgia Institute of Technology Atlanta, GA
B.S. in Industrial Engineering | Concentration: AI/OR for Decisions | Minor: Computing and Theory *May 2029*

University of Illinois Urbana-Champaign Urbana, IL
Grainger College of Engineering *September 2025 – May 2026*

Relevant Coursework: Data Science Discovery, Data Science Exploration, Calculus I–II, Linear Algebra, Discrete Math, Multivariable Calculus, Probability with Applications, Object-Oriented Programming

PROFESSIONAL EXPERIENCE

MiraPhoto *June 2026 – September 2026*
Software Engineering Intern (early-stage startup)

- Independently shipped a live Organization Profile & Event RSVP feature, recreating the UI from a screenshot
- Migrated from Next.js to Astro for static hosting; set up GitHub Actions CI/CD for GitHub Pages and Vercel
- Implemented role-based access for 3 user roles and 4 form validations (phone, URL, email, length) using Zod
- Connected the frontend to the live Node/Express/MongoDB API; resolved an undocumented schema mismatch

PERSONAL PROJECTS

Mean-Reversion Statistical Arbitrage System | Source Code

- Built a pairs-trading pipeline with log-price cointegration tests and Benjamini-Hochberg FDR correction
- Fit hedge ratios on training data only; backtested 2023–2024 out of sample with 10 bps transaction costs
- Identified 1 significant pair of 8 (KO-PEP, corrected $p = 0.036$); cost and yearly tests found no reliable edge

AirBnB Pricing Analysis: NYC vs. Chicago | Source Code

- Cleaned Kaggle data for NYC and Chicago; compared prices by room type (median $106 vs. $99)
- Built neighborhood price heatmaps showing premium listings in Manhattan, Brooklyn, and Chicago's Loop
- Modeled price with linear regression and KMeans clustering; low explanatory power (R^2 of 0.13 and 0.10)

LEADERSHIP & EXTRACURRICULAR EXPERIENCE

Hillsborough High School Swim Team *November 2021 – February 2025*
Varsity Swimmer; Team Captain (2024–2025)

- Led and mentored teammates during practices and meets, helping maintain a supportive team environment
- Competed on varsity for four years; selected team captain for the 2024–2025 season

Community Swim *June 2022 – August 2024*
Co-Founder

- Co-founded a summer swim program teaching swimming and water safety to 15 children ages 4–12
- Developed individualized swim lessons tailored to skill levels; 10 students later joined a swim team

TECHNICAL SKILLS & INTERESTS

Programming Languages: Python, TypeScript, JavaScript
ML / Data: scikit-learn, pandas, NumPy, statsmodels, matplotlib, Jupyter
Web / Tools: Node.js, Express, MongoDB, React, Astro, Next.js, REST APIs, Git
Interests: Swimming, Cycling, Gym, Gaming, Fishing, Grilling, Music, TV